

An unconditional 0.3827090879 lower bound for the union-closed sets conjecture

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Abstract

We give a computer-assisted proof that every finite union-closed family $\mathcal{F} \neq \{\emptyset\}$ contains an element belonging to at least

$$\frac{3827090879}{10^{10}}|\mathcal{F}| = 0.3827090879|\mathcal{F}|$$

members of the family. The argument uses the conditionally-i.i.d. coupling introduced by Liu, but removes the numerical structural hypotheses used for Liu’s approximately 0.38271 evaluation. The proof reduces the union-closed statement to two pointwise entropy inequalities. One is proved analytically, with one polynomial positivity step checked exactly over $\mathbb{Q}$; the other is proved by analytic boundary estimates and a rigorous two-dimensional interval certificate. The numerical component is replayed independently with directed-rounding Boost.Interval arithmetic, a separately written 256-bit MPFR implementation with an independently generated subdivision, and a non-adaptive MPFR replay of a frozen complete prefix-coded partition. This research release was generated with frontier AI; independent specialist review remains pending.

1 Introduction

The union-closed sets conjecture, usually attributed to Frankl, asks whether every nontrivial finite union-closed family contains an element present in at least half of its member sets. Gilmer obtained the first absolute constant lower bound by an entropy argument. Sawin, Yu, and Cambie subsequently developed dependent-coupling refinements reaching an unconditional frontier near 0.3823455. Liu introduced conditionally-i.i.d. couplings and proved an analytic strict improvement. A nine-dimensional numerical optimization in Liu’s paper suggested a value near 0.38271 under explicitly numerically verified structural hypotheses.

R010 replaces those hypotheses, for the exact rational constant below, by a direct two-variable inequality with a rigorous interval certificate.

Theorem 1 (R010). *Let $\mathcal{F} \neq \{\emptyset\}$ be a finite union-closed family. Then some element belongs to at least*

$$c|\mathcal{F}|, \quad c = \frac{3827090879}{10000000000} = 0.3827090879$$

members of $\mathcal{F}$.

The number is deliberately slightly below Liu’s numerically optimized value. That slack makes the decisive compact inequality strictly positive and permits a modest exact interval certificate. The theorem does not prove the 1/2 conjecture, and no optimality is claimed for c or for this proof architecture.

2 Notation and exact constants

Write

$$h(t) = -t \log t - (1-t) \log(1-t), \quad 0 \leq t \leq 1,$$

with natural logarithms, and define

$$J(x, y) = xy(1 + (1-x)(1-y)), \quad j(x) = J(x, x) = x^2(1 + (1-x)^2).$$

Use the exact constants

$$\alpha = \frac{11249343}{12500000}, \quad \beta = \frac{1250657}{12500000}, \quad c = \frac{3827090879}{10000000000},$$

$$m = 1 - c = \frac{6172909121}{10000000000}, \quad \kappa = \frac{1}{2m} = \frac{5000000000}{6172909121},$$

so $\alpha + \beta = 1$.

3 Two pointwise inequalities

The proof is driven by two statements.

Lemma 1 (Kernel inequality). *For every $x, y \in [0, 1]$,*

$$h(J(x, y)) \geq \sqrt{h(j(x))h(j(y))}.$$

Lemma 2 (Main entropy inequality). *For every $x, y \in [0, 1]$,*

$$\alpha h(xy) + \beta \sqrt{h(j(x))h(j(y))} \geq \kappa(xh(y) + yh(x)).$$

The kernel inequality is analytic apart from an exact rational polynomial positivity check. The main inequality is analytic on two boundary strips and rigorously interval-certified on the remaining compact rectangle.

4 Deduction of the union-closed bound

Let Z be uniformly distributed on the indicator vectors of $\mathcal{F}$. The case $|\mathcal{F}| = 1$ is immediate, so assume $|\mathcal{F}| > 1$. For a prefix z^{i-1} write

$$p_i(z^{i-1}) = \mathbb{P}[Z_i = 1 \mid Z^{i-1} = z^{i-1}].$$

Use two sequential couplings of copies of Z . Under $\Pi^{(0)}$, the two coordinate bits are independent Bernoulli variables with the prescribed conditional means. Under $\Pi^{(1)}$, use Liu's conditionally-i.i.d. protocol (Example 5 in [5]) with $f(r) = r(1-r)$. Let $(A^{(k)}, B^{(k)})$ be the pair generated by protocol k and set $W^{(k)} = A^{(k)} \vee B^{(k)}$. Each marginal has the law of Z .

Fix coordinate i . If

$$S = p_i(A^{(k), i-1}), \quad T = p_i(B^{(k), i-1}),$$

then S, T have a protocol-independent common marginal law. Let P_i be the law of $X = 1 - S$, and put

$$M_i = \mathbb{E}X = 1 - \mathbb{P}[Z_i = 1], \quad H_i = \mathbb{E}h(X) = H(Z_i \mid Z^{i-1}).$$

For the independent protocol, X, Y are independent with law P_i , and the conditional probability that the OR bit is zero is XY . Integrating Lemma 2 gives

$$\alpha \mathbb{E}h(XY) + \beta (\mathbb{E}\sqrt{h(j(X))})^2 \geq \frac{M_i}{m} H_i. \quad (4.1)$$

For the conditionally-i.i.d. protocol, Liu's construction gives OR-zero probability $J(X, Y)$. Conditionally on its auxiliary variable U , X, Y are i.i.d. Writing $s(x) = \sqrt{h(j(x))}$, Lemma 1, conditional independence, and Jensen give

$$\mathbb{E}h(J(X, Y)) \geq \mathbb{E}[s(X)s(Y)] = \mathbb{E}[\mathbb{E}(s(X) | U)^2] \geq (\mathbb{E}s(X))^2. \quad (4.2)$$

Thus

$$\alpha \mathbb{E}h(XY) + \beta \mathbb{E}h(J(X, Y)) \geq \frac{M_i}{m} H_i. \quad (4.3)$$

The chain rule and the fact that $W^{(k), i-1}$ is a function of the paired histories imply

$$H(W^{(k)}) \geq \sum_i H(W_i^{(k)} | A^{(k), i-1}, B^{(k), i-1}). \quad (4.4)$$

If every element occurred in strictly fewer than $c|\mathcal{F}|$ members, then $M_i > m$ for every coordinate. Summing (4.3) and using (4.4) yields

$$\alpha H(W^{(0)}) + \beta H(W^{(1)}) > \sum_i H_i = H(Z) = \log |\mathcal{F}|.$$

The inequality is strict because $H(Z) > 0$, hence some $H_i > 0$. Union closure implies each $W^{(k)}$ is supported on $\mathcal{F}$, so each entropy is at most $\log |\mathcal{F}|$, a contradiction.

5 Proof of the kernel inequality

Boundary cases follow by continuity. Assume $0 < x \leq y < 1$ and define

$$\ell(t) = \log h(t), \quad R(x, y) = 2\ell(J(x, y)) - \ell(j(x)) - \ell(j(y)).$$

Since $h'' < 0$ and $h > 0$, ℓ is strictly concave, while $R(x, x) = 0$. It suffices to prove $R_y(x, y) \geq 0$. Direct expansion gives

$$J(x, y) \leq j(y), \quad 2J_y(x, y) \leq j'(y), \quad x \leq y. \quad (5.1)$$

When $j(y) \geq 1/2$, the sign and monotonicity of ℓ' together with (5.1) imply $R_y \geq 0$.

Assume now $j(y) < 1/2$. Put $G(x, y) = \ell(J(x, y))$. Since $R_y = 2(G_y(x, y) - G_y(y, y))$, it is enough to show $G_{xy} \leq 0$. With $z = J(x, y)$,

$$G_{xy} = \ell''(z)J_x J_y + \ell'(z)J_{xy},$$

and it suffices to establish

$$-\frac{z\ell''(z)}{\ell'(z)} \geq \frac{1}{1-z} \geq \frac{zJ_{xy}}{J_x J_y}. \quad (5.2)$$

For the entropy side let $a = \log((1-z)/z) > 0$. Direct calculation gives

$$-\frac{z\ell''(z)}{\ell'(z)} = \frac{1}{(1-z)a} + \frac{za}{h(z)}.$$

After multiplying by $h(z)(1-z)a$, the required difference has numerator $N = h(z)(1-a) + za^2(1-z)$. The case $0 < a \leq 1$ is immediate. For $a \geq 1$ use

$$h(z) = za - \log(1-z) \leq z \left(a + \frac{1}{1-z} \right).$$

Because $1-a \leq 0$, it remains to use $z < (e^a)^{-1}$ and $e^a \geq a + a^2$, the latter following from the Taylor expansion of e^a .

For the geometric side exact algebra gives

$$J_x J_y - (1-J) J J_{xy} = x^2 y^2 P(x, y),$$

where

$$\begin{aligned} P(x, y) = & 4x^3 y^3 - 10x^3 y^2 + 8x^3 y - 2x^3 - 10x^2 y^3 + 34x^2 y^2 - 34x^2 y + 10x^2 \\ & + 8xy^3 - 34xy^2 + 44xy - 16x - 2y^3 + 10y^2 - 16y + 7. \end{aligned}$$

Moreover

$$j \left(\frac{84}{125} \right) - \frac{1}{2} = \frac{81647}{488281250} > 0.$$

Since j is increasing, $j(y) \leq 1/2$ implies $x, y \leq 84/125$. After rescaling to the unit square, all sixteen tensor-product Bernstein coefficients of P are positive. The exact checker `verify_auxiliary_inequality.py` recomputes the polynomial identity, the endpoint comparison, and all sixteen rational Bernstein coefficients. Hence $P > 0$, proving (5.2) and Lemma 1.

6 Proof of the main entropy inequality

For $t \in (0, 1)$ define

$$u(t) = \frac{h(t)}{t}, \quad v(t) = \frac{\sqrt{h(j(t))}}{t}.$$

After division by xy , Lemma 2 is equivalent to

$$\Delta(x, y) = \alpha u(xy) + \beta v(x)v(y) - \kappa(u(x) + u(y)) \geq 0. \quad (6.1)$$

The square is split into a small-variable strip $\min(x, y) \leq 0.01$, the compact core $[0.01, 0.999999]^2$, and a near-one strip $\max(x, y) \geq 0.999999$.

6.1 Small-variable strip

For $0 < x, y \leq 1$,

$$u(xy) \geq u(x) - \log y, \quad (6.2)$$

obtained by integrating $g'(t) = -\log(1 - e^{-t})/e^{-t} \geq 1$ for $g(t) = u(e^{-t})$. For $x \leq 0.01$,

$$j(x) \geq \frac{19801}{10000}x^2, \quad j(x) \leq 2x^2 \leq \frac{1}{5000},$$

so $h(z) \geq z \log(1/z)$ and $\log 5000 > 8$ imply $v(x) > 7/2$. Since $\beta > 1/10$, it remains to prove positivity of

$$R(y) = -\alpha \log y - \kappa u(y) + \frac{7}{20}v(y).$$

Elementary entropy estimates handle $0 < y \leq 10^{-4}$ and $0.9999 \leq y < 1$. On $[10^{-4}, 0.9999]$, the rigorous interval routine `verifyR()` uses 39 boxes and has reference least lower endpoint 0.0033854057973891488.

6.2 Near-one strip

By symmetry assume $x \leq y$ and $y \geq 0.999999$, with $x > 0.01$. If $x \leq 0.998$, monotonicity of u gives a directed-rounding lower bound 0.0012884232397950039. For $0.998 \leq x \leq y < 1$, concavity of $g(t) = u(e^{-t})$ gives

$$u(xy) \geq \frac{u(x^2) + u(y^2)}{2}.$$

It is therefore enough to show $\alpha u(z^2) > 2\kappa u(z)$ for $0.998 \leq z < 1$. Writing $z = 1 - \varepsilon$ and using

$$h(t) = t(1 - \log t) - E(t), \quad 0 \leq E(t) \leq \frac{t^2}{2(1-t)},$$

reduces the claim to explicit rational/logarithmic estimates; the resulting published lower bound is $391/12500 > 0$.

6.3 Compact core

On $[0.01, 0.999999]^2$, the primary checker uses Boost.Interval with IEEE-754 directed rounding. It does not delegate logarithms to system libm. For $1 \leq q \leq 2$, with $w = (q - 1)/(q + 1)$, it encloses

$$\log q = 2 \sum_{n=0}^{24} \frac{w^{2n+1}}{2n+1} + R_{24}$$

using the explicit positive remainder bound

$$0 \leq R_N \leq \frac{2w^{2N+3}}{(2N+3)(1-w^2)}.$$

The checker combines natural interval evaluation with a second-order Taylor enclosure on each rectangle and bisects only when no current enclosure proves a nonnegative lower endpoint. The reference Boost run reports 3273 accepted core boxes, maximum depth 39, and least accepted lower endpoint $4.1293562138181227 \times 10^{-12}$.

7 Independent numerical verification

The numerical obligation is deliberately replayed through distinct trust paths. The Boost path uses directed binary64 arithmetic and an internally bounded logarithm series. A separately written MPFR checker uses 256-bit outward rounding, MPFR’s own logarithm and square-root functions, exact rational domain endpoints, and an independently generated subdivision; its reference core contains 3210 boxes and has least lower endpoint $4.1428237777820989 \times 10^{-12}$. The same MPFR subdivision is frozen as prefix-coded leaf paths. A non-adaptive static checker validates the trie as a complete partition before checking the 3210 leaves. Finally, the Bernstein step in Section 5 is recomputed with exact `fractions.Fraction` arithmetic.

Search and heuristic optimization are therefore outside the final correctness oracle. The theorem relies on the analytic reductions above and on the exact or directed-rounding certificate checks just described.

8 Reproducibility and trust boundary

From a clean Ubuntu/Debian checkout install `g++`, `python3`, `Boost`, `GMP`, and `MPFR` and run `./verify_release.sh`

The release materializer validates the SHA-256 of the compressed verifier and certificate payload and rejects unsafe archive paths and links. CI replays the exact auxiliary check, Boost certificate, independent MPFR certificate, static MPFR certificate, and public release-integrity checks.

No proof-assistant formalization is currently part of the R010 trust boundary. A future Lean formalization would be most valuable for the protocol-to-entropy deduction and the analytic reduction from the two pointwise inequalities to Theorem 1. The large interval partition is already represented as a frozen deterministic certificate checked by independent rigorous programs.

9 Relation to prior work and novelty boundary

Gilmer [1] initiated the constant entropy method. Sawin [2], Yu [3], and Cambie [4] developed dependent-coupling refinements, with an unconditional frontier around 0.3823455. Liu [5] introduced the conditionally-i.i.d. protocol used here and proved an analytic strict improvement. Liu’s numerical section reported an approximately 0.38271 value under numerically verified hypotheses arising from a nine-dimensional optimization.

R010 does not claim invention of the protocol or of the numerically suggested target. Its contribution is the unconditional exact rational constant 0.3827090879, obtained by replacing those structural/global-optimization hypotheses with the two pointwise inequalities proved and rigorously certified here.

10 Limitations

The full Frankl $1/2$ conjecture remains open. This work does not establish the best constant obtainable by entropy methods, and it does not prove Liu’s full numerically suggested decimal 0.382709087918741... Independent specialist review remains pending.

References

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- [5] J. Liu, *Improving the Lower Bound for the Union-closed Sets Conjecture via Conditionally IID Coupling*, arXiv:2306.08824v1, 2023.